

Kazi Reyazul Hasan

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Research Focus

- Language Processing
- Speech & Gesture
- Behavioral Modeling
- Multimodal AI

Education

Bangladesh University of Engineering and Technology Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering Feb 2020–Mar 2025
CGPA: 3.72/4.00 (Last 4 semesters: 3.88/4.00, 2x Dean's List)
Thesis: *MediaPipe-Aided Ensemble Learning with Attention-Based Transformers for Word-Level Sign Language Recognition*

Research Experience

Bangladesh University of Engineering and Technology Dhaka, Bangladesh
Research Assistant Dec 2023–Present
Supervisors: Dr. Muhammad Abdullah Adnan, Dr. A. B. M. Alim Al Islam

- **Sign Language Recognition:** Developed EMPATH framework achieving 91.07% accuracy on Bangla word-level sign language recognition, surpassing previous state-of-the-art by 5–8% through novel MediaPipe integration with attention-based transformers. Published at ICPR 2024 and RRPR 2024 (oral presentations).
- **Writing Biometrics:** Proposed a training-free authorship pipeline using LLM-generated stylometric profiles over frozen sentence encoders for verification and identification. Accepted at IJCB 2026.
- **Speech Emotion Recognition:** Designed dual-stream DNN-KAN architecture with Bangla-specific phonetic features, achieving 95.35% accuracy with only 1.1M parameters on Bangla speech emotion recognition. Accepted at Interspeech 2026.
- **Technical Domain Translation:** Created BanglaSTEM, the first technical domain parallel corpus for Bangla-English translation with 5,000+ sentence pairs across mathematics, physics, and programming. Accepted at ACL SRW 2026.
- **Visual Speech Recognition:** Built BanglaLRS, the first continuous open-vocabulary Bengali visual speech recognition benchmark (100h, speaker-disjoint), with a Conformer baseline reaching 36% CER / 57% WER (vs. 52% CER for Arman-AV, Persian, 220h). Under review at WACV 2027.

Publications

Peer-Reviewed Conference Papers.....

[1]: **EMPATH: MediaPipe-Aided Ensemble Learning with Attention-Based Transformers for Accurate Recognition of Bangla Word-Level Sign Language.**
Kazi Reyazul Hasan and Muhammad Abdullah Adnan
Oral Presentation at International Conference on Pattern Recognition (ICPR 2024) | Springer | Paper

[2]: **Toward Training-Free Writing Biometrics: LLM Stylometric Profiles for Authorship Verification and Identification.**
Kazi Reyazul Hasan, A. B. M. Alim Al Islam, and Muhammad Abdullah Adnan
Accepted at International Joint Conference on Biometrics (IJCB 2026)

[3]: **Dual-Stream DNN-KAN Networks with Bangla-Specific Features for Speech Emotion Recognition.**
Kazi Reyazul Hasan and Muhammad Abdullah Adnan.
Accepted at Interspeech 2026 Main Track

Workshop Papers.....

[4]: **BanglaSTEM: A Parallel Corpus and Term-Weighted Evaluation for Technical Bangla-English Translation.**
Kazi Reyazul Hasan, A. B. M. Alim Al Islam, and Muhammad Abdullah Adnan
ACL Student Research Workshop 2026 | Paper

[5]: **Enhancing Word-Level Sign Language Recognition.**
Kazi Reyazul Hasan and Muhammad Abdullah Adnan
Oral Presentation at Reproducible Research in Pattern Recognition (RRPR 2024) | Springer | Paper

[6]: **TF-IDF Informed Prompt Engineering with LoRA Fine-tuning for Bangla Hate Speech Detection.**
Kazi Reyazul Hasan et al.
Shared Task Paper, 2nd BLP Workshop, IJCNLP-AAACL 2025 | Paper

Under Review.....
[7]: BanglaLRS: A Continuous, Open-Vocabulary Benchmark for Bangla Visual Speech Recognition.
Kazi Reyazul Hasan et al.
Under review at IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027), Evaluations & Datasets track

Professional Experience

- Presidency University
Adjunct Lecturer (Part-time)

Dhaka, Bangladesh
Jan 2026–Present

 - Taught 200+ students CSE courses including Introduction to Computer Programming, Object Oriented Programming, Digital Logic Design, Computer Networks and Security.
- TigerIT Bangladesh Limited
Software Engineer, R&D

Dhaka, Bangladesh
Jul 2025–Present

 - Achieved **1st position** on the FVC-onGoing Differential Morphing Attack Detection (DMAD) benchmark with an AdaFace ResNet FR backbone and a lightweight detection head, reaching 0.00% EER on DMAD-MORPHDB_D-1.0 and 0.19% EER on DMAD-MORPHDB_P&S-1.0.
 - Developing deep learning-based iris recognition system using Vision Transformers and circular CNNs to replace traditional Hamming distance methods, targeting sub-1% Equal Error Rate (current system ranks 14th globally on IREX 10).
- Robofication LLC
Junior AI Engineer (Remote)

Michigan, United States
Mar 2025–May 2025

 - Designed multimodal system modeling interface using JointJS, FastAPI, and React for SysML diagram generation.
 - Developed LLM-based strategies for artifact exchange between proprietary AI toolchain and Cameo Systems Modeler.

Academic Service

2026: Reviewer, AIMS Workshop, CVPR 2026
2026: Program Committee Member, Reproducible Research in Pattern Recognition (RRPR 2026), ICPR Workshop

Grants & Awards

Jul 2026: Travel Grant, Interspeech 2026, Sydney, Australia
Aug 2025: CEO, SignSpeak - Selected for BUET Incubator Program
Sep 2024: Research Grant, ICT Innovation Fund of Government of Bangladesh
Apr 2024: Internal Research Grant, BUET Research and Innovation Centre for Science and Engineering (RISE)
Mar 2023: Best Project Award, CSE-316, BUET

Featured Projects

EMPATH: TensorFlow, Keras, Transformers, MediaPipe, OpenCV, Algorithms | **Code, Slide**
Developed a sign language recognition framework that exceeds previous accuracy benchmarks by 5–8%.

Busbuddy: Node.js, Svelte, PostgreSQL, AWS | **Code**
Led the development of the BUET Bus Automation service admin interface.

Race: C++, Arduino | **Code, Award**
Constructed a fully functional hardware racing game simulation with two custom-built cars.

Leadership & Service

Academic Team Member, Bangladesh Mathematical Olympiad (5+ years);
Leadership roles including Joint Secretary across multiple student organizations;
50+ awards in olympiads, quizzes, and debate competitions

Technical Skills

ML / NLP: PyTorch, TensorFlow, Transformers (Hugging Face), OpenCV, MediaPipe, scikit-learn, Langchain, Embedding Models, Prompt Engineering, LoRA Fine-tuning
Systems / Lang.: PostgreSQL, Firebase, AWS, Azure, FastAPI, Docker, Git; Python, C++, Java, JavaScript